CTMT Atlas and Null-Sector Information

Orbit–Stabilizer Classification, Čech Globalization, and the Exact Information Content of the Unresolved Sector

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Abstract

This paper is the global companion to the resolution-geometry foundation $\mathcal{G} = (O, g; R, \Sigma)$ and its covariance layer. It consolidates the mathematically defendable content of the CTMT collapse, holonomy, coupling, and atlas programme, aligned to the metric-explicit formulation. Three results are established. First, an *exact information theorem* for the unresolved sector: the null sector carries, on its own, zero first-order information about the parameters, yet whenever it is covariance-coupled to the resolved sector it strictly increases the resolved Fisher information, the increase being exactly the Schur-complement term (Theorem 3.3). Second, an orbit–stabilizer classification of resolution geometries under their automorphism groups, giving a well-defined notion of observable similarity type. Third, a Čech globalization in which admissible transitions take values in the structure group $\mathrm{Aut}(\mathcal{G})$, weak globalization is ordinary cocycle gluing, and holonomy is the obstruction to trivialization rather than a failure of gluing. We are explicit about the boundary of these results: they are *static and second-order* (correlational) statements. The stronger reading of coupling as a *dynamical* accumulation—"buildup by data leak"—is not expressible in the present static formalism and is stated as the central open problem (Section 7), together with the dynamical structure a genuine buildup theorem would require.

Contents

1 Purpose and claim stratification

The foundation isolates the local object $\mathcal{G} = (O, g; R, \Sigma)$ with resolved sector $R = \mathrm{Im}(J)$, null sector $N = R^{\perp_g}$, and covariance Σ; the covariance paper studies its block structure $(\Sigma_R, \Sigma_N, C_{RN})$ and the singular case. This paper is the global and information-theoretic companion. We separate, as before, what is proved from what is interpreted.

Proved (this paper).
The exact information contribution of the null sector (Theorem 3.3); orbit–stabilizer classification (Section 4); Čech globalization with structure group $\mathrm{Aut}(\mathcal{G})$ and holonomy as trivialization obstruction (Section 5).

Interpreted, not proved.
That coupling represents a dynamical flow or accumulation from unresolved to resolved directions. The static formalism does not express accumulation; see the explicit boundary in Section 7.

Non-claim 1.1 (Static, not dynamical)*.* All coupling results below are second-order statistical statements about a single operating point: one Jacobian J and one covariance Σ. They do not describe evolution in time and do not, by themselves, establish any accumulation, cascade, or causal transfer between sectors. Terminology such as "leakage" is used only in the precise sense $C_{RN} \neq 0$.

2 Objects recalled

We use the conventions of the foundation: (O, g) is the observation space with fixed reference metric (whitening $g = \Sigma^{-1}$ when $\Sigma \succ 0$), $N := R^{\perp_g}$, and g-orthogonal projectors P_R, P_N. In an adapted g-orthonormal frame

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^* & \Sigma_N \end{pmatrix}, \qquad \Sigma_R = P_R \Sigma P_R, \quad \Sigma_N = P_N \Sigma P_N, \quad C_{RN} = P_R \Sigma P_N.$$

An admissible morphism is a g-isometry L with $L(R_1) = R_2$ and $L\Sigma_1 L^* = \Sigma_2$ (whence $L(N_1) = N_2$); the automorphism group is $\mathrm{Aut}(\mathcal{G}) = \{L \in \mathrm{O}(g) : L(R) = R,\ L\Sigma L^* = \Sigma\}$, characterised by the coupling-stabilizer theorem of the foundation: $L = \mathrm{diag}(A, B)$ with $A\Sigma_R A^* = \Sigma_R$, $B\Sigma_N B^* = \Sigma_N$, and $AC_{RN}B^* = C_{RN}$.

3 The exact information content of the null sector

This is the rigorous core of CTMT's claim that the unresolved sector is not disposable. It is a static, second-order theorem; it is exactly the classical statement that correlated but signal-free observations are informative. It is verified numerically.

Assumption 3.1 (Observation model)**.** Let $y \sim \mathcal{N}(\mu(\theta), \Sigma)$ with $\Sigma \succ 0$ and $J = \partial\mu/\partial\theta$ satisfying $\mathrm{Im}(J) = R$. In an adapted frame $J = \binom{J_R}{0}$, so the mean varies only in the resolved sector: $P_N \mu$ is constant in θ.

Lemma 3.2 (The null sector is signal-free on its own)**.** *Under Assumption 3.1, the observation restricted to the null sector, $y_N = P_N y \sim \mathcal{N}(P_N \mu, \Sigma_N)$, has Fisher information about θ equal to zero.*

Proof. The mean of y_N is $P_N\mu$, constant in θ by Assumption 3.1, so $\partial(P_N\mu)/\partial\theta = 0$ and the Gaussian Fisher information $(\partial_\theta P_N \mu)^\top \Sigma_N^{-1} (\partial_\theta P_N \mu) = 0$. □

Theorem 3.3 (Null-sector information contribution). *Under Assumption 3.1, let*

$$\mathcal{I}_R := J_R^\top \Sigma_R^{-1} J_R \quad (\textit{resolved-only observer}), \qquad \mathcal{I}_{\text{full}} := J^\top \Sigma^{-1} J \quad (\textit{full observer}).$$

Then

$$\mathcal{I}_{\text{full}} = J_R^\top S_R^{-1} J_R, \qquad S_R := \Sigma_R - C_{RN}\Sigma_N^{-1} C_{RN}^\top \ (\textit{Schur complement}),$$

and

$$\boxed{\mathcal{I}_{\text{full}} \ \succeq \ \mathcal{I}_R}, \qquad \mathcal{I}_{\text{full}} - \mathcal{I}_R = J_R^\top \big(S_R^{-1} - \Sigma_R^{-1}\big) J_R \ \succeq 0.$$

The inequality is strict on $\operatorname{Im}(J_R^\top)$ *if and only if* $C_{RN} \neq 0$.

Proof. $\mathcal{I}_{\text{full}} = J^\top \Sigma^{-1} J = J_R^\top (\Sigma^{-1})_{RR} J_R = J_R^\top S_R^{-1} J_R$ by block inversion. Since $C_{RN}\Sigma_N^{-1} C_{RN}^\top \succeq 0$, we have $S_R \preceq \Sigma_R$, and inversion on the positive-definite cone is order-reversing, so $S_R^{-1} \succeq \Sigma_R^{-1}$ and hence $J_R^\top S_R^{-1} J_R \succeq J_R^\top \Sigma_R^{-1} J_R$. Equality holds iff $S_R = \Sigma_R$, i.e. iff $C_{RN}\Sigma_N^{-1} C_{RN}^\top = 0$; as $\Sigma_N \succ 0$ this is iff $C_{RN} = 0$. □

Corollary 3.4 (The defensible CTMT null-sector statement). *The unresolved sector carries no first-order signal in isolation (Lemma 3.2), yet observing it strictly improves inference of the resolved parameters exactly when it is covariance-coupled to the resolved sector (Theorem 3.3). The magnitude of the improvement is the Schur-complement information gain* $J_R^\top (S_R^{-1} - \Sigma_R^{-1}) J_R$.

Remark 3.5 (Direction of the effect, stated carefully). The gain is a gain in *information* about resolved parameters obtained by a full observer relative to a resolved-only observer. Equivalently, the resolved-block *conditional* covariance shrinks: $S_R \preceq \Sigma_R$. This is the standard nuisance/conditioning identity and is the precise, correct content behind statements that "the null sector feeds the resolved sector." It is informational and static; see Non-claim 1.1 and Section 7.

4 Orbit–stabilizer classification

Since $\operatorname{Aut}(\mathcal{G})$ is a closed subgroup of the compact group $\mathrm{O}(g)$, it is a compact Lie group acting smoothly on O; orbit–stabilizer theory applies without qualification.

Definition 4.1 (Similarity, orbit, stabilizer). For $x \in O$, the similarity orbit and stabilizer are $\operatorname{Orb}(x) = \{Lx : L \in \operatorname{Aut}(\mathcal{G})\}$ and $\operatorname{Stab}(x) = \{L \in \operatorname{Aut}(\mathcal{G}) : Lx = x\}$.

Theorem 4.2 (Orbit–stabilizer). *Each orbit is a homogeneous space,* $\operatorname{Orb}(x) \cong \operatorname{Aut}(\mathcal{G})/\operatorname{Stab}(x)$, *and* $\dim \operatorname{Orb}(x) = \dim \operatorname{Aut}(\mathcal{G}) - \dim \operatorname{Stab}(x)$.

Proof. The action map $L \mapsto Lx$ is a smooth surjection $\operatorname{Aut}(\mathcal{G}) \to \operatorname{Orb}(x)$ whose fibres are the left cosets of $\operatorname{Stab}(x)$; the induced bijection $\operatorname{Aut}(\mathcal{G})/\operatorname{Stab}(x) \to \operatorname{Orb}(x)$ is a diffeomorphism for a compact Lie group acting smoothly. The dimension formula follows. □

Definition 4.3 (Similarity type). The similarity type of $\mathcal{G}$ is the isomorphism invariant

$$\tau(\mathcal{G}) = \big(\dim O,\ \dim R,\ \operatorname{spec}\Sigma_R,\ \operatorname{spec}\Sigma_N,\ \sigma(\Sigma_R^{-1/2} C_{RN} \Sigma_N^{-1/2}),\ \operatorname{Aut}(\mathcal{G})\big),$$

where the fifth entry is the canonical-correlation spectrum of the whitened coupling and all entries are $\mathrm{O}(g)$-invariant.

Proposition 4.4 (Coupling refines the classification). *Two geometries with equal* $\dim O, \dim R$ *and equal* $\operatorname{spec}\Sigma_R, \operatorname{spec}\Sigma_N$ *may still be non-isomorphic: if their whitened-coupling canonical correlations differ, no admissible morphism relates them, and their automorphism groups may differ (foundation, coupling-stabilizer theorem). Hence* C_{RN} *is a genuine classification invariant, not a removable term.*

Proof. Canonical correlations are admissible-morphism invariants (they are $\mathrm{O}(g|_R) \times \mathrm{O}(g|_N)$ invariants of C_{RN}); distinct values obstruct isomorphism. The automorphism-group dependence is the coupling stabilizer $\{(A, B) : A\Sigma_R A^* = \Sigma_R,\ B\Sigma_N B^* = \Sigma_N,\ AC_{RN}B^* = C_{RN}\}$. □

5 Čech globalization

Globalization is the standard Čech gluing problem with one CTMT-specific feature: the structure group is $\mathrm{Aut}(\mathcal{G})$ rather than a general linear group. The gluing theorem is not new mathematics; the content is the identification of the structure group and the correct placement of holonomy as an obstruction, not a failure. This section supersedes the "lock"-style statements of the source drafts (see Remark 5.7).

Definition 5.1 (Atlas and transitions)**.** An atlas is a cover $\{(U_\alpha, \mathcal{G}_\alpha)\}$ with, on each nonempty overlap, an admissible morphism $\phi_{\alpha\beta} : \mathcal{G}_\beta \to \mathcal{G}_\alpha$ satisfying $\phi_{\alpha\alpha} = \mathrm{id}$ and $\phi_{\beta\alpha} = \phi_{\alpha\beta}^{-1}$. Transitions take values in $\mathrm{Aut}(\mathcal{G})$ on self-overlaps and, in general, in the admissible-isomorphism set between charts.

Definition 5.2 (Coherent atlas / cocycle condition)**.** An atlas is *coherent* if on every nonempty triple overlap

$$\phi_{\alpha\beta} \circ \phi_{\beta\gamma} = \phi_{\alpha\gamma}.$$

Theorem 5.3 (Weak globalization)**.** *A coherent atlas glues to a global resolution-geometry bundle with structure group* $\mathrm{Aut}(\mathcal{G})$*, unique up to isomorphism induced by* 0*-cochain gauge transformations* $\phi_{\alpha\beta} \mapsto a_\alpha \phi_{\alpha\beta} a_\beta^{-1}$*.*

Proof. Take the disjoint union of local trivial pieces and identify points across overlaps by the $\phi_{\alpha\beta}$. Reflexivity and symmetry are $\phi_{\alpha\alpha} = \mathrm{id}$ and $\phi_{\beta\alpha} = \phi_{\alpha\beta}^{-1}$; transitivity on triple overlaps is Definition 5.2. The quotient carries local trivialisations whose transitions are the $\phi_{\alpha\beta}$, valued in $\mathrm{Aut}(\mathcal{G})$. This is the standard fibre-bundle gluing construction with structure group reduced to $\mathrm{Aut}(\mathcal{G})$. □

Corollary 5.4 (Pairwise admissibility is insufficient)**.** *Admissibility of every overlap map does not imply gluing; the triple-overlap cocycle condition is an independent hypothesis.*

Definition 5.5 (Holonomy)**.** For a closed chart chain $\gamma = (\alpha_0, \dots, \alpha_p = \alpha_0)$, $\mathrm{Hol}_\gamma = \phi_{\alpha_0\alpha_1} \cdots \phi_{\alpha_{p-1}\alpha_p} \in \mathrm{Aut}(\mathcal{G}_{\alpha_0})$. The holonomy group is $\mathrm{Hol}(\alpha_0) = \{\mathrm{Hol}_\gamma\} \le \mathrm{Aut}(\mathcal{G}_{\alpha_0})$.

Theorem 5.6 (Holonomy is the trivialization obstruction, not a failure)**.** *For a coherent atlas with flat (locally constant) transitions, a global trivialisation over the cover exists if and only if the cocycle is a coboundary, equivalently iff the holonomy representation is trivial up to gauge. Nontrivial holonomy does not obstruct existence of the glued bundle; it classifies nontrivial global geometries.*

Proof. A global trivialisation is a family a_α with $\phi_{\alpha\beta} = a_\alpha^{-1} a_\beta$, i.e. the 1-cocycle is a coboundary. For flat transitions the same data is the monodromy/holonomy representation of loops into $\mathrm{Aut}(\mathcal{G})$; triviality up to gauge is exactly existence of the a_α. A nontrivial class still defines a bundle via Theorem 5.3; it merely cannot be globally trivialised. □

Remark 5.7 (On the removed "lock" theorems)**.** The source drafts state a "collapse–holonomy lock" and a "leakage–instability lock" asserting that "either bounded admissible trivialization exists or it does not." As stated these are instances of the excluded middle and carry no mathematical content beyond their definitions; we therefore do not present them as theorems. Their useful residue is the *diagnostic taxonomy* of Section 6: a named, ordered list of closure checks whose first failure localises the obstruction. That taxonomy is an engineering convenience, not a theorem, and is presented as such.

6 Diagnostics and falsifiers

The following are computational checks, ordered so that the first failure names the obstruction. They are diagnostics, not theorems.

Check	Quantity	Failure meaning
Rank	$\lvert \operatorname{rank} J_\alpha - \operatorname{rank} J_\beta \rvert$	resolved-dimension change
Sector	$\lVert P_{R,\alpha} - \phi P_{R,\beta} \phi^{-1} \rVert$	sector drift
Covariance	$\lVert \phi \Sigma_\beta \phi^* - \Sigma_\alpha \rVert$	covariance not preserved
Coupling	canonical-correlation mismatch	different similarity stratum
Information	$\lVert S_R^{-1} \rVert$ conditioning	Schur-driven ill-conditioning
Cocycle	$\lVert \phi_{\alpha\beta}\phi_{\beta\gamma} - \phi_{\alpha\gamma} \rVert$	incoherent atlas
Holonomy	Hol_γ class	nontrivial global class (not failure)

Falsifier 6.1 (Coupling stratum mismatch)**.** If two charts share $\dim R$, $\operatorname{spec} \Sigma_R$, $\operatorname{spec} \Sigma_N$ but differ in whitened-coupling canonical correlations, they are not admissibly isomorphic (Proposition 4.4); a transition claiming otherwise is rejected.

Falsifier 6.2 (Incoherent atlas)**.** If pairwise transitions are admissible but the triple-overlap cocycle residual exceeds tolerance, no coherent atlas exists on that cover without refinement.

Falsifier 6.3 (Fisher-only false positive)**.** Equality of Fisher pullbacks does not imply equality of covariance geometry (covariance paper, Schur insufficiency); Fisher agreement alone must not be accepted as chart compatibility.

7 Scope boundary: what a genuine "buildup" theorem would require

The results above are static and correlational. The stronger reading of the null sector as a source of *dynamical accumulation*—the informal "buildup by data leak"—is not a theorem of this framework, and cannot be, because the present objects contain no dynamics: a single Jacobian and a single covariance describe one operating point. We state precisely what would be needed, so that the claim is a well-posed target rather than a slogan.

Open Problem 7.1 (Dynamical resolution geometry)**.** Equip the resolution geometry with an evolution: a one-parameter family $\mathcal{G}(t) = (O, g; R(t), \Sigma(t))$ generated by an observable transfer/evolution operator Φ_t on O (e.g. $\dot{y} = Ay +$ noise with stationary or Lyapunov covariance $\Sigma(t)$). Only in such a model is "accumulation" meaningful.

Open Problem 7.2 (Buildup as monotone information transfer)**.** Within a dynamical model (Problem 7.1), formulate "buildup" as a monotone quantity, e.g. a nondecreasing null$\to$resolved directed information or transfer entropy, or a sign condition on $\frac{d}{dt} C_{RN}(t)$ coupled to growth of the resolved covariance. Prove conditions on (A, Σ) under which this quantity is strictly increasing. Absent such a theorem, "buildup" remains an interpretation of the static gain of Theorem 3.3.

Open Problem 7.3 (Nonstatic-rank globalization)**.** Extend Section 5 to families where $\operatorname{rank} J$ or $\operatorname{range} \Sigma$ changes across charts, i.e. singular globalization (covariance paper, singular-support results): classify the resulting stratified obstructions.

Non-claim 7.4 (Honest status of the thesis). This paper proves that the unresolved sector is informative for resolved inference under correlation (Theorem 3.3) and that it refines the global classification (Proposition 4.4). It does *not* prove that unresolved variability dynamically accumulates into the resolved sector. That statement requires the dynamical layer of Problems 7.1–7.2, which is not developed here.

8 Conclusion

The defendable global content of CTMT is now a coherent package. The unresolved sector is not noise to be discarded: it carries zero first-order signal alone, yet strictly sharpens resolved inference whenever coupled, by a computable Schur-complement amount (Theorem 3.3). Resolution geometries are classified up to admissible isomorphism by an orbit–stabilizer similarity type in which the coupling canonical correlations are a genuine invariant (Section 4). Globally, charts glue by Čech cocycles with structure group $\mathrm{Aut}(\mathcal{G})$, and holonomy classifies nontrivial global geometries rather than signalling failure (Section 5).

The one thing this framework does not yet do is the thing its informal name promises: a static geometry cannot exhibit accumulation. The precise dynamical structure required to turn the correlational gain into a genuine buildup theorem is stated as Problems 7.1–7.2. Making "buildup by data leak" a theorem—rather than an interpretation of Theorem 3.3—is the natural next paper, and the honest place to draw the current boundary.

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